

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5CJ301			
Course Title	REAL ANALYSIS II			
Type of Course	Major			
Semester	V			
Academic Level	300 – 399			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	3	2	75
Pre-requisites	1. Mathematical Logic and necessary exposure to set theory. 2. Basic Calculus 3. Real Analysis I			
Course Summary	Continuous real functions are introduced rigorously using the epsilon-delta argument. The equivalent sequential criterion is established later. Differentiable and (Riemann) Integrable functions are introduced followed by the fundamental theorem of calculus connecting the two notions. The course concludes with a discourse on series of functions and various results discussing the compatibility of the above three notions with the limiting operations on series of functions.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse and explain the concept of continuous functions and their properties on intervals, and apply the principles of uniform continuity.	An	C	Internal Exam/Assignment/ Seminar/ Viva/Report/ End Sem Exam
CO2	Analyse the vitality of continuous functions when they are defined on intervals.	An	C	Internal Exam/Assignment/ Seminar/ Viva/Report/ End Sem Exam
CO3	Apply the derivative and the Mean Value Theorem to solve problems and prove related theorems.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva/Report/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Introduction to Real Analysis, 4/e, Robert G Bartle, Donald R Sherbert John Wiley & Sons(2011)			
Module	Unit	Content	Hrs (45+30)	Marks Ext:70
I	Continuous Functions		14	Min.15
	1	Section 5.1 – Continuous functions		
	2	Section 5.3 – Continuous functions on intervals — 5.3.1 to 5.3.5		
	3	Section 5.3 – from 5.3.7 - 5.3.10		
	4	Section 5.4 – Uniform Continuity-up to 5.4.3		
	5	Section 5.4 – Uniform Continuity-5.4.4 to 5.4.14(proof of Weierstrass Approximation Theorem is optional)		
	6	Selected problems from the above sections.		
II	Differentiation		10	Min.15
	7	Section 6.1 – The Derivative – 6.1.1 to 6.1.7		
	8	Section 6.2- The Mean Value Theorem - 6.2.1 to 6.2.6		
	9	Section 6.2 - from 6.2.7 to 6.2.9		
	10	Section 6.2-The Mean Value Theorem- 6.2.10 to 6.2.13		
	11	Selected problems in the above sections.		
III	The Riemann Integral		14	Min.20
	12	Section 7.1 – Riemann Integral – up to 7.1.4 (a)		
	13	Section 7.1 – from 7.1.5 to 7.1.7 (proof of 7.1.7 is optional)		
	14	Section 7.2 – Riemann Integrable functions – 7.2.1 to 7.2.5 (Examples 7.2.2 are optional)		
	15	Section 7.2 – from 7.2.7 to 7.2.13		
	16	Section 7.3 – The Fundamental Theorem – 7.3.1 to 7.3.7		
	17	Section 7.3 – from 7.3.8 to 7.3.18 (proof of theorem 7.3.18 is optional)		
	18	Selected problems in the above sections.		
IV	Sequences and Series of functions		7	Min.10
	19	Section 8.1 – Pointwise and Uniform Convergence – 8.1.1 to 8.1.3		
	20	Section 8.1 – from 8.1.4 to 8.1.10		
	21	Section 8.2 – Interchange of limits – 8.2.1		
	22	Section 8.2 – Interchange of limit and continuity - 8.2.2		
V	Practicum: The goal is for the students to learn the following selected topics in 15 practicum sessions of two hours each via self-study and group activities. The lecturer should assist them by running group discussions, overseeing class seminars and referring library books for self-study and note preparation.		30	
	1	Section 5.2 – Combinations of continuous functions		
	2	Section 5.6 – from 5.6.5 to 5.6.7		

	3	Section 6.1 – Inverse Functions – 6.1.8 to 6.1.10		
	4	Section 6.3 – L'Hospital's Rule -from 6.3.5 to 6.3.7		
	5	Section 6.4 – Taylor's theorem – 6.4.1 to 6.4.4		
	6	Section 8.2 – Interchange of Limits – 8.2.3 and 8.2.4		
	7	Section 9.1 – Absolute Convergence – 9.1.1 to 9.1.3		
	8	Section 9.1 – 9.1.4 to 9.1.5		
	9	Section 9.2 – Limit Comparison Test with examples		
	10	Section 9.2 – Root Test with examples		
	11	Section 9.2 – Ratio Test with examples		
	12	Section 9.2 – Integral Test with examples		
	13	Section 9.2 – Raabe's Test with examples		
	14	Section 9.3 – Alternating Series Test		
	15	Section 9.4 – Infinite Series – Series of Functions – 9.4.1 to 9.4.7		

Reference

1. Apostol, Tom M. Calculus, Volume 1. John Wiley & Sons, 1991.
2. Tom.M. Apostol, Mathematical Analysis, 2/e, Addison-Wesley, 2002.
3. Richard R Goldberg, Methods of Real Analysis, 2/e, Wiley, 2020
4. Raymond L Wilder, Introduction to the Foundations of Mathematics, 2/e, John Wiley & Sons
5. Malik, Subhash Chandra, and Savita Arora. Mathematical analysis. New Age International, 1992.

***Optional topics are exempted for end semester examination**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	2	1	2	0	2	0	3	0	0
CO 2	2	2	2	1	2	0	2	0	3	0	0
CO 3	3	2	3	1	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓